

Week 3: BERTweet Integration and ML Classification

ML Service

The third week focused on integrating the machine-learning component into RESQAI.

A separate **Python FastAPI service** was created to handle ML inference.

The ML service contains:

```
```text
```

```
ml-service/
```

```
■■■ app/
```

```
■ ■■■ services/
```

```
■ ■ ■■■ bertweet_service.py
```

```
■ ■ ■■■ gemini_service.py
```

```
■ ■■■ main.py
```

```
■■■ models/
```

```
■■■ requirements.txt
```

```
■■■ .env
```

```
The BERTweet service loads the trained model and performs classification on incoming disaster-related
```

```
BERTweet Classification
```

```
BERTweet was selected as the primary classification model.
```

```
The model returns:
```

```
```text
```

```
Category
```

```
Confidence Score
```

The system supports ten categories, including:

- requests_or_urgent_needs
- injured_or_dead_people
- missing_or_found_people
- displaced_people_and_evacuations
- infrastructure_and_utility_damage
- rescue_volunteering_or_donation_effort
- caution_and_advice
- other_relevant_information

- sympathy_and_support
- not_humanitarian

Confidence-Based Routing

A confidence threshold of **0.70** was established.

The classification logic works as follows:

```
```text
```

```
IF confidence >= 0.70
```

```
AND category != "not_humanitarian"
```

↓

Use BERTweet result

```
Otherwise, the message is sent to Gemini for additional analysis.
```

```
Example
```

```
For a message such as:
```

```
```text
```

```
"We urgently need water and medical supplies"
```

BERTweet may produce:

```
```text
```

```
Category: requests_or_urgent_needs
```

```
Confidence: 0.92
```

```
Since the confidence is above 0.70, the BERTweet result is accepted without calling Gemini.
```

```
Outcome
```

```
By the end of Week 3, the primary ML classification pipeline was integrated with the backend, allowing
```